

CIVIL 3D – PART BUILDER PART 2

Posted by John Evans | Nov 25, 2008 | AutoCAD Civil 3D

Source : <https://designandmotion.net/autodesk/autocad-civil-3d/civil-3d-part-builder-part-2/>

Introduction

Today we continue from where we left off in Part Builder Part 1.

Before we proceed, I'd like to note that a wing wall may not be everyone's biggest desire. I needed it, and I felt that it would provide a fairly well rounded view of things. It needs equations, and gets worked laterally instead of vertically, which make it more of a challenge.

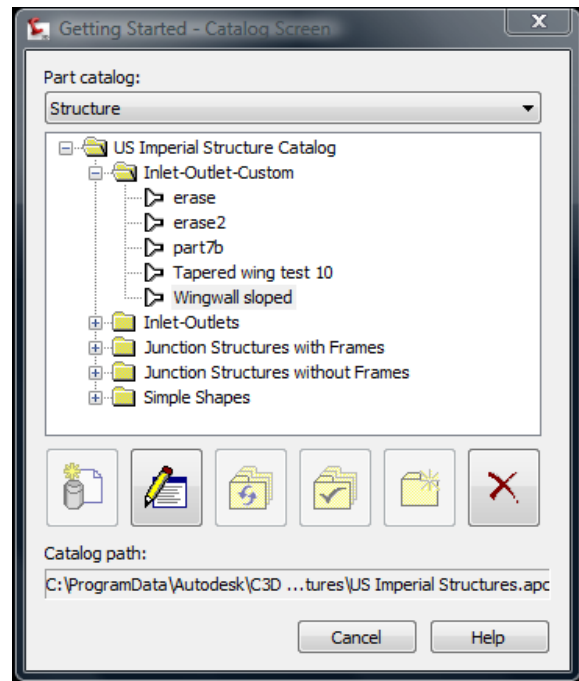
Sitting on the axis of 2 different careers I can see this type of workflow from 2 different perspectives, and want to provide everyone with a bit more information that may have been unavailable.

- The 3D modelers understand the construction methods. What they need to understand is the limitations stopping them from working naturally through this process.
- The Civil 3D techs know about odd application limitations, but need to understand how the construction methods, dependencies, and calculations work.

If you don't get something out of this session, it will be because you already had a good working knowledge of Part Builder, or you failed to post your questions here.

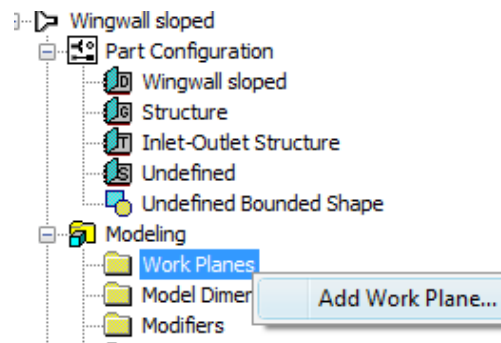
The 3d Model (which is what you came for, right?) is created using the following procedures: Create a Work Plane, Create Geometry, Add Dimensions, Add Constraints, Create Profile, Modify Dimensions, Create Modifier, and Configure Parameters. We will discuss the first six today.

Open the part for the last session, named Wingwall Sloped.

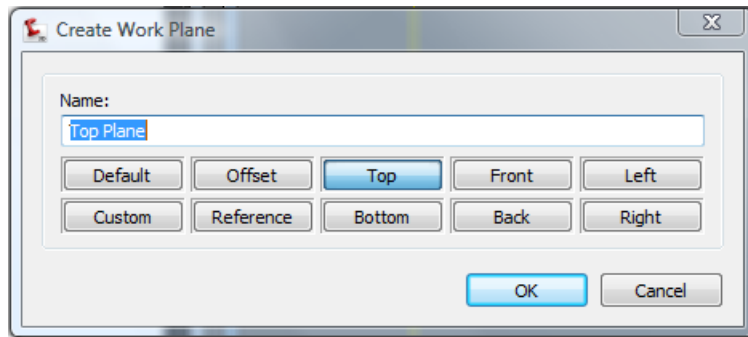


CREATE A WORK PLANE

Let's begin by creating the Top Work Plane. I always like to get my Top plane in first. Either I am working on it, or visually referencing it. Under Modeling, Right Click the Work Planes folder and select 'Add Work Plane...'.



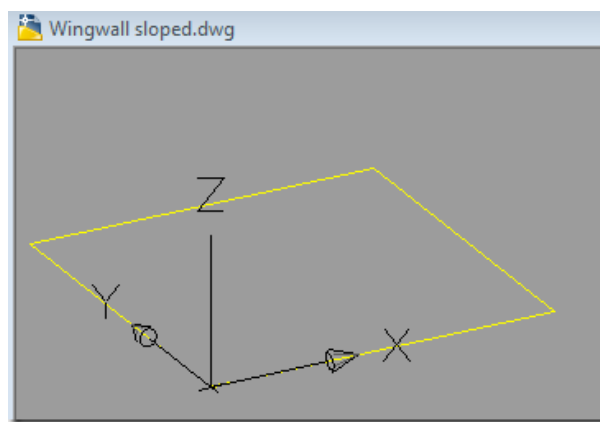
The Create Work Plane dialog will appear, and allow us to select 1 of 10 different options. Select the Top Plane button, and the application will default the name to be Top Plane. You can change this to be anything you like. I like Top Plane.



After we are finished with the name, click 'OK'. The work plane is created, and it shows up in the model space view to the right. Let's rotate the view a bit so we can get some perspective.

Hold the Shift key down while pressing the Pan button on your mouse. (Alternately you can navigate through the menus: View menu->3D Orbit->Free Orbit, or enter 3dorbit at the command line).

Gently push up until the plane begins to flatten, and then pull to the right. You should see something like the image below.



This Yellow square is a visual representation of the Top Plane. There are no limits. The 'Box' you see is just to help your mind grasp the location of the plane in 3D space. Remember our last discussion. Top plane is King, and the structure drag leader always seems to point to the X axis related to the Top plane. This means that if we build something, the pipe will naturally want to come in from the Right side of our image. It just happens that that is called the Right Plane. We need that plane created as well.

Once again, Right Click the Work Planes folder and select 'Add Work Plane...'. Select the Right Plane button, and the application will default the name to be Right Plane.



Notice the UCS axis change. Remember that each plan has its own ucs. The one to keep your eyes on is related to the Top plane.

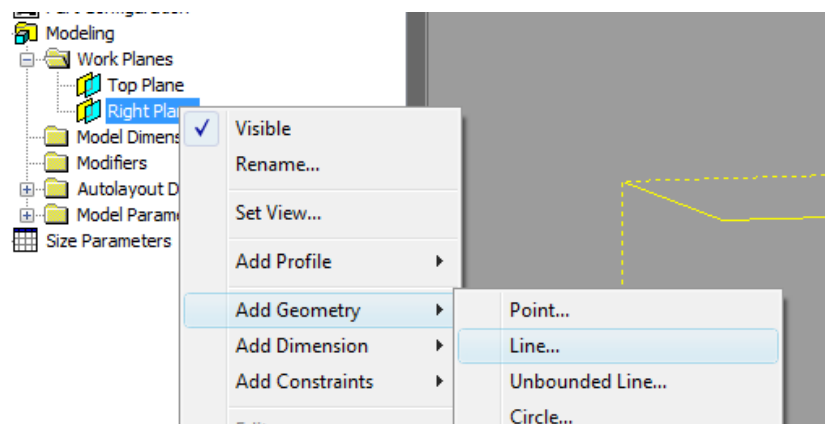
WORK PLANE COLLECTIONS

Now that we have a work plane created, features can be added. Each feature, such as Geometries created and subsequently associated to a work plane will be cataloged in a tree view. As you need, simply expand the respective plane, and navigate/expand the collection headings to view and modify the features, constraints, and dimensions.

Lines, Points, Circles, etc. are accessed under the Geometry headings. Dimensions and Constraints are cataloged under their respective headings. Profiles are cataloged directly under their respective work planes, not with the geometries. More about profiles shortly.

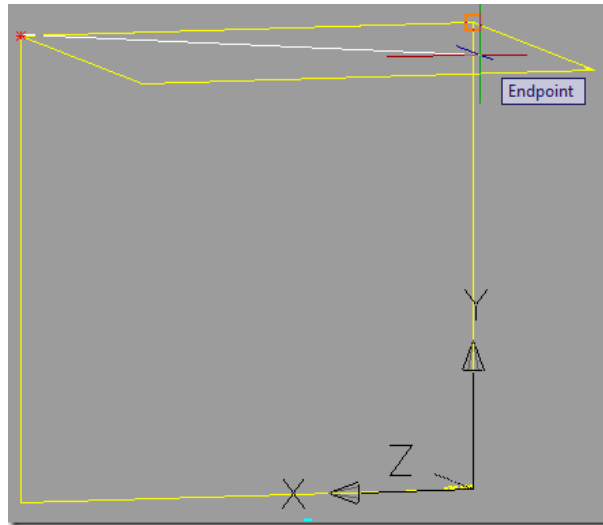
CREATE GEOMETRY

Let's create some geometry. Rotate the view so that the Right Plane is in front of you. Right click on the Right Plane, and select Geometry, and then Line...



*[*TIP: Using Endpoint Osnap, we can Pick the intersecting corners of the Top/Right Planes. This will set the corner of the structure on both the Right and Top planes.]*

This little cheat will save about 3 minutes of work, and a lot of headaches down the road. You will not have mismatched features on perpendicular planes (requiring additional planes to align the connections; Conversely you will not be required to construct temporary geometry on 1 plane to reference to the other, and then have to erase the original.



Lines are segments that lie between 2 points created on the similar plane. The points actually control the line. Without 2 points, the line cannot exist. Every feature will be controlled by the respective point feature(s).

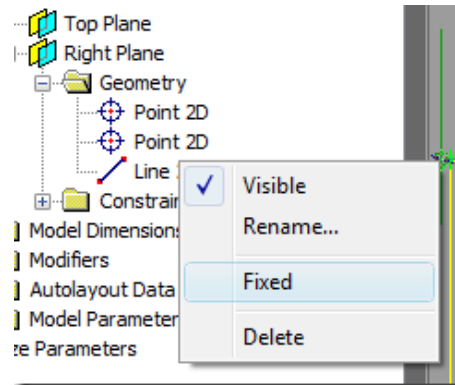
Now a word about anchors and dimensions. The Part Builder acts like Inventor in the fashion that features are dimensioned. In Civil and AutoCAD, the item is drawn using references, offsets, and properties to size it to suit application. Then dimensions are applied to display the resulting feature size. In Part Builder (and Inventor) the size of the feature is controlled BY THE DIMENSIONS. Once the user applies the dimension, the value can be edited, and the feature will update accordingly.

We need to anchor the part down so we can control how the feature changes. Once you draw a feature, it will stay in place as long as nothing (such as dimensions) affects it. We will be affecting it a lot.

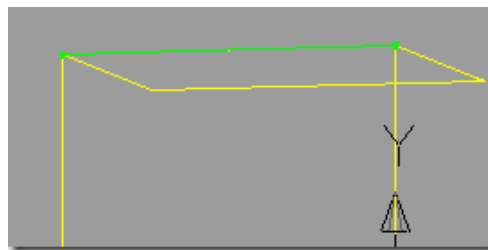
To illustrate this, let's say you create a rectangle approximately 12"x12". You would like it to be 24" square. You would need to apply a dimension to control the exact size. After you apply the dimension of 24", and the feature decides to update, which side moves? Ponder that one for awhile.

We need to anchor the feature by applying its Fixed flag.

Under the Right Plane, expand Geometry, and Right click on each of the features that you previously created. Select 'Fixed'.

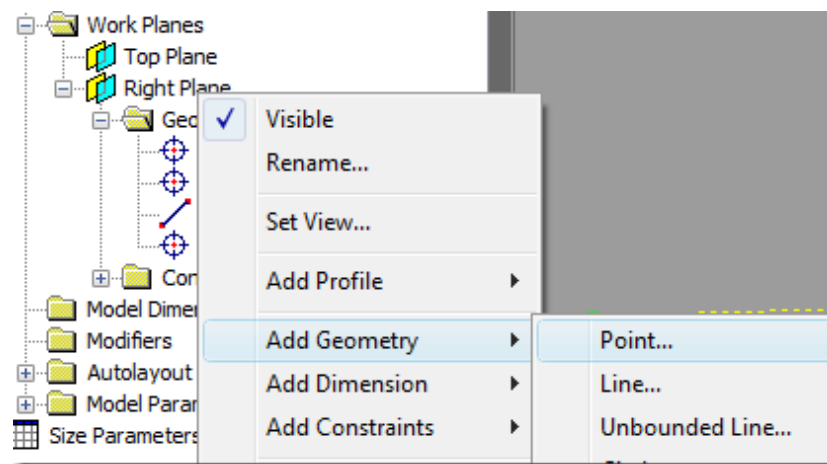


Each item that has been 'Fixed' will turn green, indicating that it cannot move. Could I have just fixed the 2 end points? Yes, but having all 3 fixed is a potential benefit later, and the visual reference from the line is helpful as well.

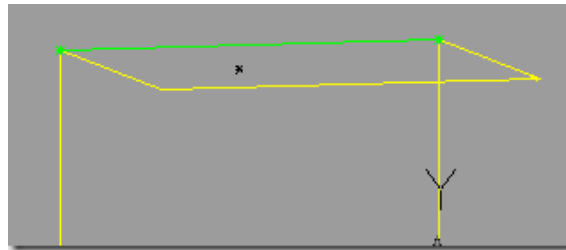


We need something to represent the centerline of the part. We should set a point to fix in space that will later be used as a reference.

Right click on the Right Plane, and select Geometry, and then point...



We need to set the point to be constrained at the midpoint of our anchor. Pick on a location that is similar to the one in the image below. I always set modeling features AWAY from the intended location, so I can be sure that the constraints are controlling the object properly.



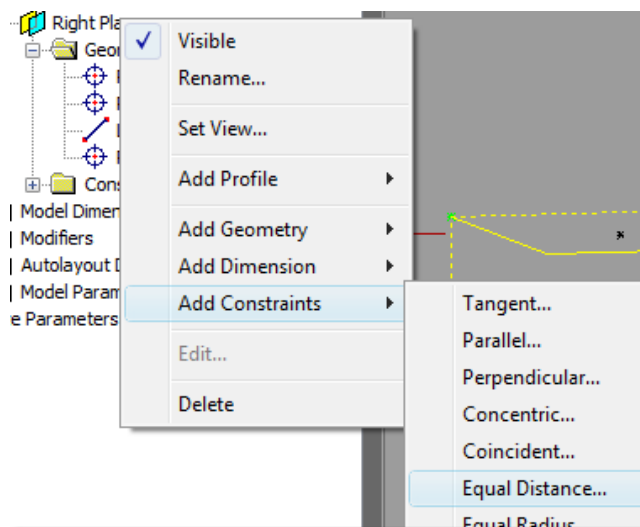
ADD CONSTRAINTS

Constraints are like rules that specific geometries must follow, like behavioral boundaries. Unlike my children, the application will not ignore its rules. In AutoCAD, the technician indicates what changes are applied, and how they should be applied. In a parametric 3D model, the 3D solid will expand and contract according to the parameters defining it. Without constraint, there is no knowing how the modeled parts will interact with each other. Conflicts would crop up everywhere. With constraints, we can tell a feature to consistently be parallel to another, no matter what. As the solid grows, the application knows that it has to evaluate the constraints, and then the changes. Let's get this thing under control.

Right click on the Right Plane, and select Add Constraints, and then Equal Distance...

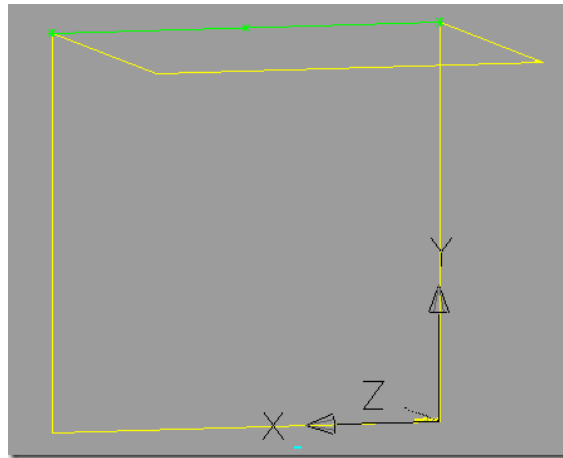
The application will ask us to pick 2 sets of geometry to compare.

Select the new Point, then the Green Point to the right. Now select the Green point at the left, and then the new point again.



The application will make the distances match (with no specific constraint about where it leaves it though). Now we need to constrain it ON the green line. Return through the Add Constraints menu, and select Coincident...

Select the new point, and then the green line. The point will land on the line, constrained equally distant from each end point of the green line.



We have now created the anchor and alignment reference for the Right Plane.

Note: A point can only be used for geometry on the same plane. Just because a Point lies at the intersection of 2 planes does not mean it can be shared between them. If you need to create features on different planes that intersect, duplicate points must be created at the intersection on the different planes.

Now we need the feature that will represent the face of the wall. This will be the point from which all lateral dimensions are referenced.

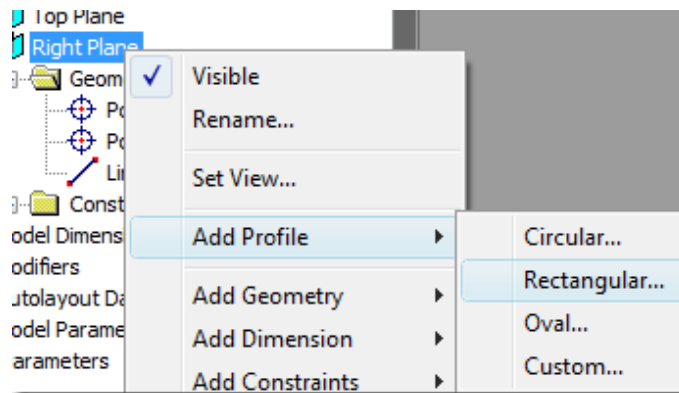
CREATE A PROFILE

Profiles are combined and named geometries, like a group of lines or a polyline. Think of them as the final product feature, resulting from various combinations of the items in the remaining groups (such as Geometries, and Dimensions). *The application creates a 3D solid from Profiles, not from geometries.*

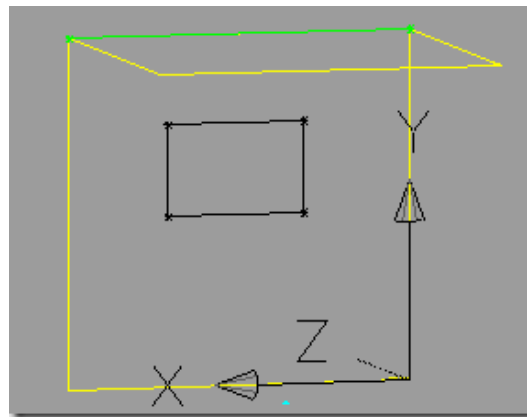
There are 2 ways to get the profile created:

- [Add Profile->Custom] Select various geometries and add them to a custom Profile, like a point, and add lines to a polyline.
- [Add Profile->Rectangular] Alternately, you can go directly to a preset Profile that automates the geometry creation, like typing the rectangle command. The perpendicular constraints are also created as well. This is the method I will use whenever possible.

Right click on Right Plane, Select Add Profile, and then Select Rectangular...



Pick two corners in the middle of the yellow plane reference.



In Inventor, there is a saying that helps new modelers get features created with more stability. “constrain first, dimension last”. Oddly enough, this will be the complete opposite in part builder.

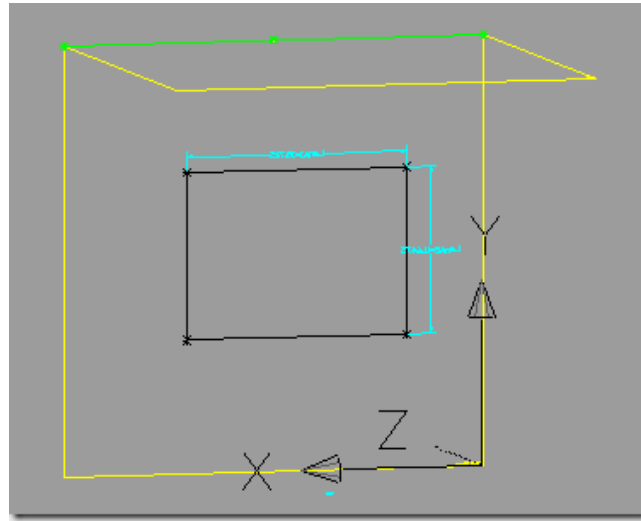
“DIMENSION FIRST, CONSTRAIN LAST!!!!”

This saying applies to the management of each individual feature. There is nothing to stop you from applying other constraints and features later, but initially, this is a good rule of thumb. The worst thing that can happen if you don’t is for geometries to begin overlapping because there was no dimension between them. You can’t just pick and move them back.

ADD DIMENSIONS

Right click on Right Plane, Select Add Dimension, then Distance...

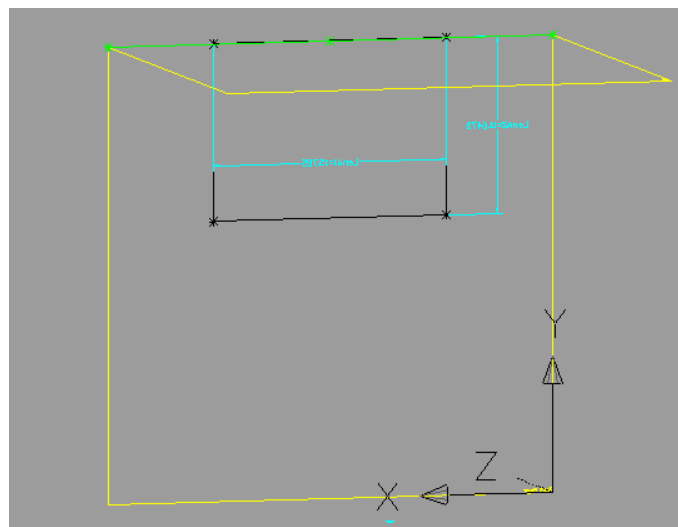
Select the 2 points at the top of the new rectangle. Position the dimension above the feature. Repeat the process for the 2 points at the right side of the rectangle.



We can modify the dimensions whenever we chose. For now, let's constrain the feature into place. We need the rectangle to be equally spaced across the middle green point, and coincident to the green anchor line. Right click on the Right Plane, and select Add Constraints, and then Equal Distance...

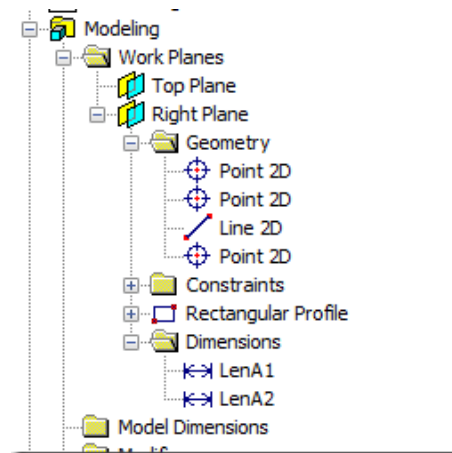
Select the Center green point, and the upper right point on the rectangle, and then the point at the upper left corner of the rectangle, and the center green point again. The rectangle will move, possible rotate, but will not change shape. We also need to get it constrained to the anchor line to stop it from flailing about. Return through the Add Constraints menu, and select Coincident...

Select the upper line of the rectangle, and then the green anchor line. The image below shows the result.



The feature has been constrained, and distributed, and is a very good foundation for the rest of the part. Look at the Part Browser. You should notice by now the part features distributed under the 'Right Plane' collection.

In the Geometry grouping you should notice the 3 points and the line created. The Profile is a separate item, and the 2 dimensions are found under Dimensions. When ever something needs to be adjusted, or a constraint is becoming problematic, simply navigate to the item and right click it.



Right click on the Dimension 'LenA2' (the vertical dimension). Select 'Set Value'. On the command line, you should see the prompt "Enter a Distance". Enter 30.

The feature should grow dramatically taller than what it was. Save the part.

I hope that you learned a little along the way today. The most important lessons I think covered for someone new is Constraints, Anchoring, Finding your features in the Browser, and "Dimension First, Constrain Last".

You can overdo constraints pretty easy. If the feature gets over-constrained, or behaves oddly, just use some judgement, and delete constraints that are suspect. Practice will help with this judgement over time.

One other warning from Autodesk is that everything you use is pulling memory for a little part. Work efficiently. If you need 2 lines that connect, you only need 3 points, not 4. Work planes are also something to be conserved. Don't worry, the simple part we will create is not going to break our budget.

In the next session we will cover User parameters, Extrusions, Transitions, and some equations too. We'll also cover a strategy with the User Parameters to buffer catastrophes if parameters get fouled up. Read the next post in [Part Builder Part 3](#).